Integrated investigation

A holistic study of a lake ecosystem would involve looking at all the inputs into the lake and its margins as well as all the outputs, including energy, water, minerals, and nutrients. It would also consider any human inputs. The study would examine the roles played by both producer organisms, such as plants and algae, and consumers such as herbivores and carnivores. The holistic approach also examines changes over time, in which developments that benefit some organisms in the short-term might lead to a lack of diversity in the future. For example, although trout thrive in warmish, alkaline waters, if those waters become too warm or acidic due to ecological change, the fish can no longer breed.

Odum’s holistic approach leaves a legacy of a far more detailed appreciation of what is happening in an ecosystem than a series of individual species studies.